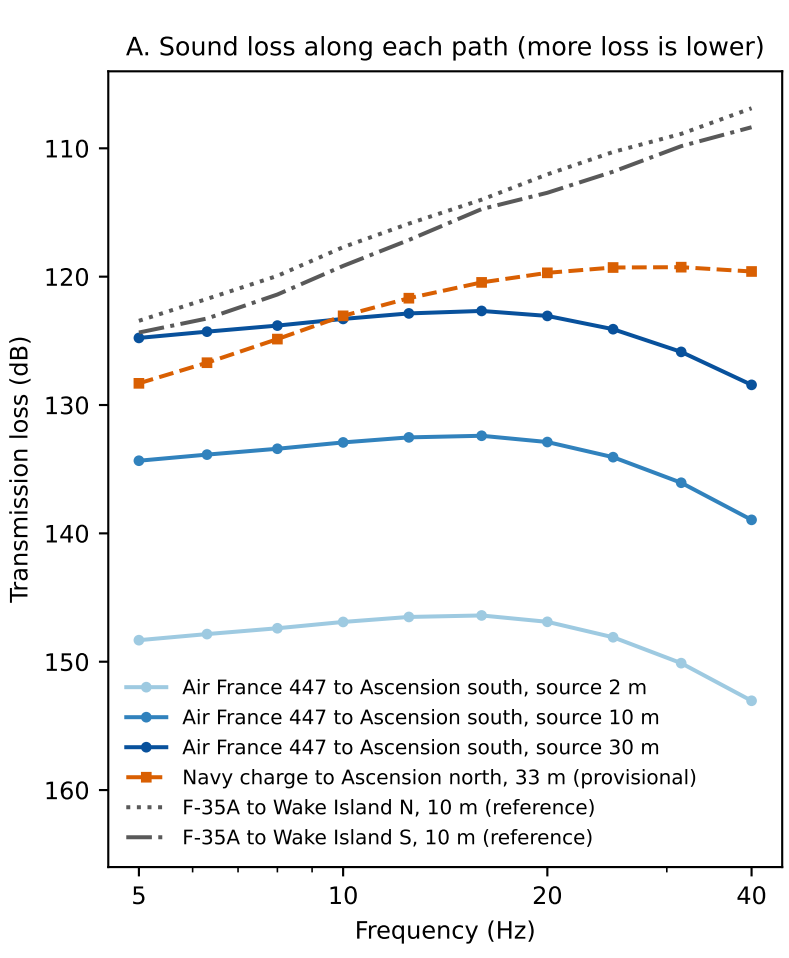
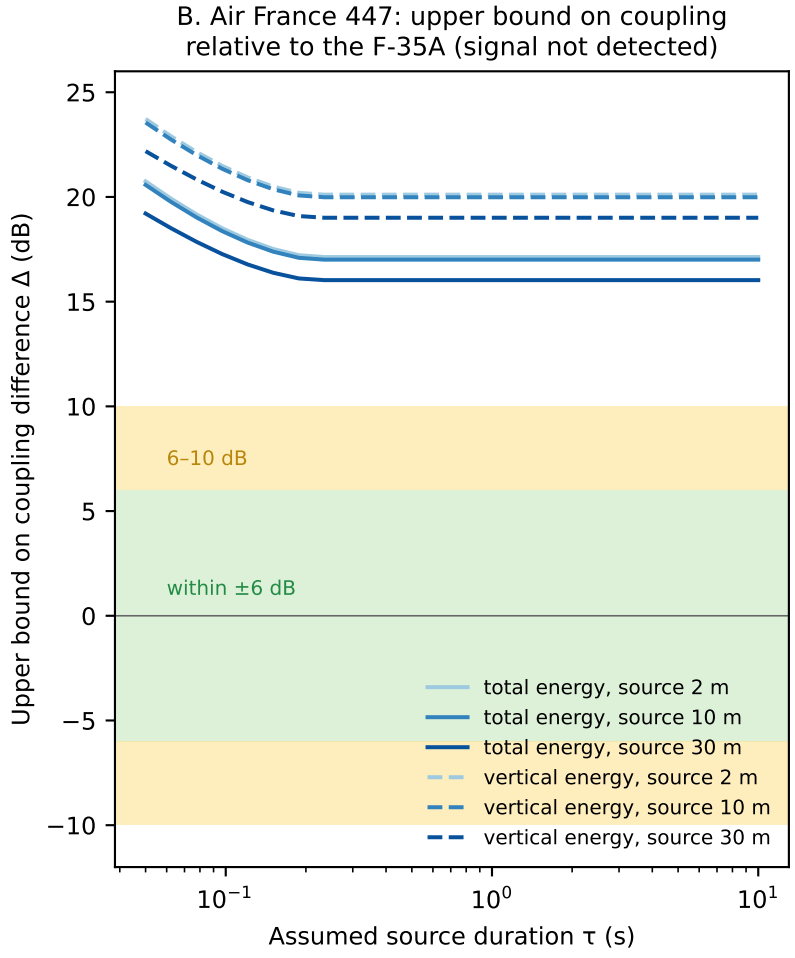
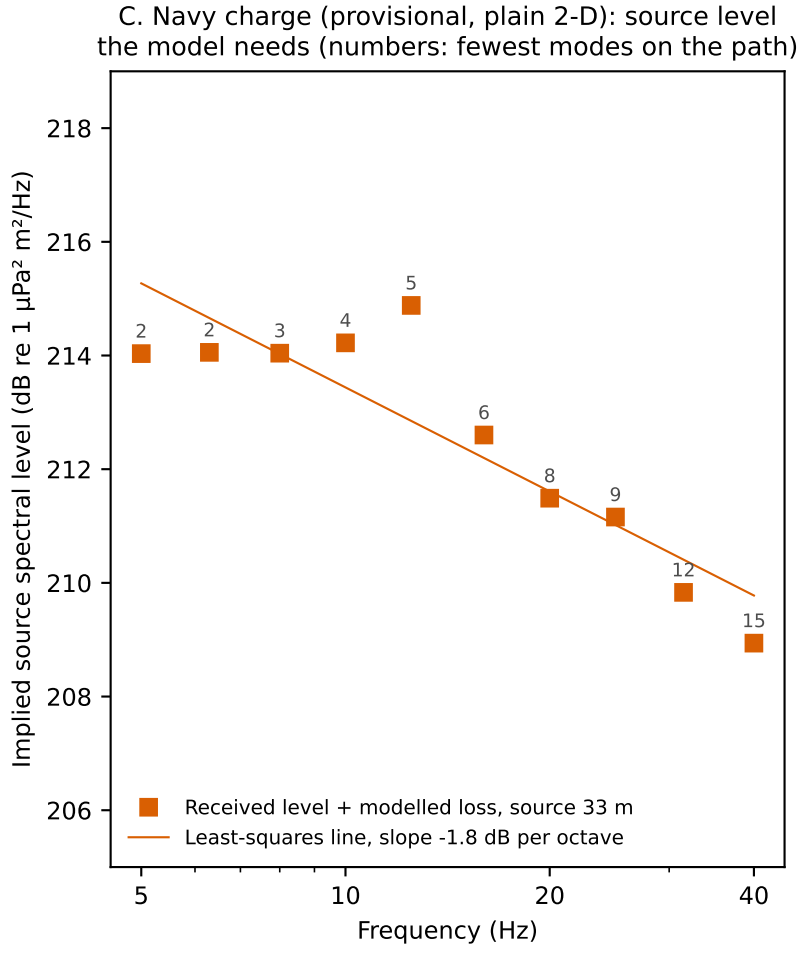
Calibration events, stage 1 (stand-in run of the module's recipe). Not a likelihood. Run C sensitivity in the same note is UNCONVERGED.







STE100: Panel A shows how much sound the model loses between each event and the hydrophone. Panel B: the Air France 447 signal was not seen above the noise in the published figure, so the panels give only a maximum. If the crash put the same part of its energy into sound as the F-35A, the model predicts a peak of about 0.13 Pa, which is below the noise. To see a peak, the crash must put about 20 dB (100 times) more of its energy into sound than the F-35A. So this event does not test the ±6 dB or ±10 dB limits. Panel C: a ridge (Rio Grande Rise) is on the path; the result is not final.

Technical: KRAKEN adiabatic normal modes, incoherent sum, hard fluid half-space (cp 1650 m/s, ρ 1.9, 0.5 dB/λ), Francois–Garrison, spherical correction, third-octave 5–40 Hz; GEBCO_2026 3×3 median + WOA23 (A5B4 June; B5C2 Dec), ocean-transport sections dafb0d1. B: Δ_ratio = 10log10[(p_e/p_F)²(E_F/E_e)G_F/G_e] (prereg 23add102 am. 1), p_e = 0.914 Pa (Kadri 2024 Fig. 5a, SNR 1.32, upper limit), p_F = 1.15/1.22 Pa (Fig. 4a, mean of H11N/H11S Δ), E_F = 900 MJ, E_e = 625 MJ total / 315 MJ vertical (BEA), band fraction slope 2, τ grid 0.05–10 s. On 86% of the AF447 path the critical depth is absent or below the seabed, so the TL depends on the bottom model. C: Nielsen et al. 2021 Fig. 5c (module digitisation), noise subtracted in power; PSD ordinate, so the source unit needs Nielsen's window before an absolute check. Rise at 2,590–2,672 km (track 674 m, channel axis ~850 m): needs N×2D; H04S path not yet built, so the pre-registered station difference is not computable.